

DEVALEKKA PALANIVEL

AI & Data Science | Aspiring Data Analyst | Power BI | Python | Machine Learning
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SUMMARY

Data-driven problem solver with hands-on experience in ML, data analysis, and visualization. Published researcher in deep learning for fraud detection. Top 500 idea (out of 24,411) at Niral Thiruvizha, Tamil Nadu's state-level innovation competition.

TECHNICAL SKILLS

ML/AI: Python, Scikit-learn, Random Forest, LightGBM, CatBoost, CNN, BiLSTM, Attention Mechanism, NLP, MediaPipe, OpenCV

Analytics: Power BI, Excel, Pandas, NumPy, Matplotlib, Seaborn, EDA, Web Scraping

Database: SQL

Deployment: Github, Gradio, Hugging Face Spaces, Streamlit,

RESEARCH PUBLICATIONS (IJSART)

A Hybrid Deep Learning Model for Credit Card Fraud Detection (CNN-BiLSTM + Attention) [\[Paper\]](#) Apr 2026
Achieved 99.32% accuracy (F1: 97.29%) on European dataset; 98.67% accuracy (F1: 95.80%) on Indian dataset. Deployed via Streamlit with SHAP explainability.

A Survey on ML & DL Approaches for Credit Card Fraud Detection [\[Paper\]](#) Feb 2026
Reviewed CNN, BiLSTM, MLP, Autoencoders, and Ensemble models addressing class imbalance in fraud detection.

INTERNSHIPS

Data Science & Machine Learning Intern | Upskill Campus Jul – Aug 2025

- Built a Turbofan Engine RUL prediction model using sensor data; applied Python, data preprocessing, and ML algorithms.

Data Analytics Intern | CodeAlpha May – Jun 2025

- Developed a web scraping pipeline (Python + BeautifulSoup), performed EDA and visualized insights with Matplotlib.

PROJECTS

NutriSnap – AI Nutrition Tracker | Python, ML, NLP, Mobile App | Niral Thiruvizha Top 500 / 24,411

- AI-powered mobile app that tracks daily nutrition via food photo, voice, or text input using a CV model + standard nutrition database.
- Features night-based streak tracking, rewards, and friend leaderboards for engagement; supports Tamil & English with voice interaction.
- Selected among Top 500 out of 24,411 ideas at Niral Thiruvizha, Tamil Nadu's state government innovation competition; advancing to next stage.

YogaMate – Real-Time AI Yoga Pose Coach | Python, MediaPipe, OpenCV

[GitHub](#)

- Real-time yoga coaching app with live pose estimation for 20+ poses, voice feedback, wrong-pose detection, and a hold timer.

Turbofan Engine RUL Predictor | Python, Random Forest, LightGBM, CatBoost, Gradio

[GitHub](#)

- Stacked ensemble model on 21 sensor readings to predict remaining engine cycles for predictive maintenance; deployed on Hugging Face.

Carbon Estimator AI | Python, Random Forest Regressor, Gradio, Hugging Face

[Live Demo](#)

- ML-powered carbon footprint estimator ($R^2 = 0.91$) deployed on Hugging Face with visual breakdowns and eco tips.

EDUCATION

B.Tech in Artificial Intelligence & Data Science

Nov 2022 – May 2026

Chettinad College of Engineering and Technology, Karur | CGPA: 8.1 (Current)

CERTIFICATIONS

Frontend Technologies — IBM (Oct 2024)

Business Intelligence & Analytics — NPTEL (Apr 2025)

Data Fundamentals — IBM (Jun 2025)

Data Science Tools — IBM (Jun 2025)